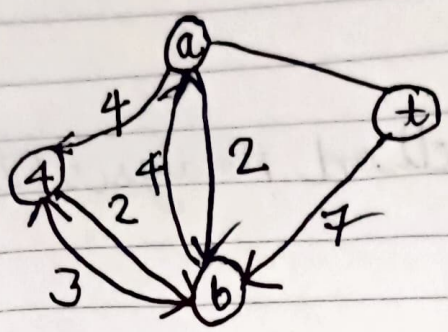
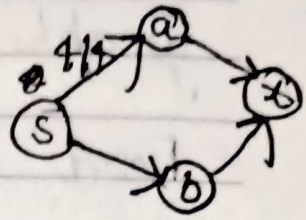
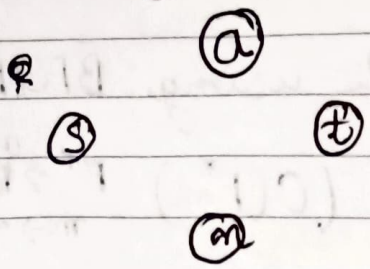
10/09/24

Algorithm (PB)







flow value stops if there is no path

Lemma 1: Let f & f' are flows in G and G^f then, $f+f'$ (flow) & $|f|+|f'|$ (val of flow) is a flow in G

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Pf: Since f and f' are both valid flows, which satisfies capacity constraint & flow convervation $|f|+|f'|$ will satisfy the capacity and

flow conservation property in original graph

Ford Fulkerson Method Analysis (time complexity)

1) finding s-t path using BFS/DFS

$O(n+m) \approx O(m)$ $(O(F^* m)$ $F^* \to$ amt of flow, $m \to$ no of edge

2) How many times s-t path to find the maximum value of flow $|f| = F^*$

3) it is a pseudopolinomial time complexity.

Cuts:

observation

1) Flow cant increase from even

2) Some of the edges got saturated ($f(u,v) = c(u,v)$)

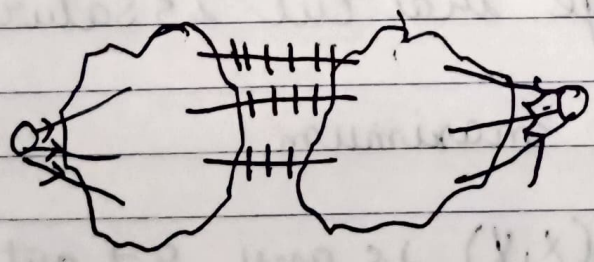
3) Not all vertices are reachable from s

4) There is a partition of vertices

X: vertices that are reachable from s

Y: vertices that are not reachable from s.

$\sum c(u,v)$ upper bound of value of flow



Def of cut:

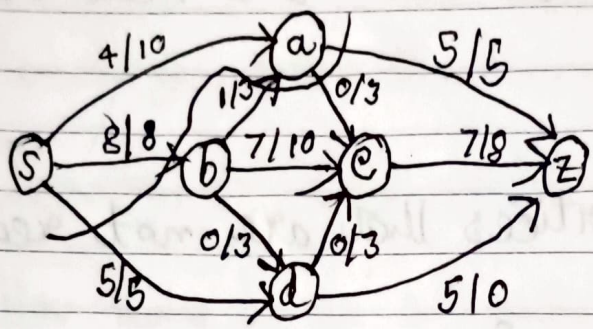
Given a network G, an s-t cut is a partition of vertex into two disjoint

$X \xrightarrow{24} Y$

subsets $X \subseteq V$ & $Y = V \setminus X$ where $s \in X$, $t \in Y$

Def^n: net flow from X to Y is sum of $f(X,Y) = \sum_{x \in X} \sum_{y \in Y} f(x,y) - \sum_{y \in Y} \sum_{x \in X} f(y,x)$

Obs^n: $f(x,y) - f(Y,X)$



$X = \{s, a\}$

$Y = \{b, c, d, t\}$

$f(X,Y) = 5 + 0 + 8 + 5 - 1$

$= 17$

if each edge that cut is saturated then it is maximum

Lemma: Let (X,Y) is any s-t cut in a network. Given any flow f, $|f| = f(X,Y)$

$|f| = f^{out}(S) - f^{in}(S)$ [$f^{in}(S)$ is 0

for all other nodes of X except s satis,

fies flow conservation

Thus $|f| = \sum_{x \in X} [f^{out}(x) - f^{in}(x)]$

obsn: For every edge (u,v) if u&v both are in X.

For every edge (u,v) if u&v both are in X,

$\sum f(u,v)$

$u,v \in X$ - i) contribution

$u \in X, v \in Y$ - ii)